

1. Find the volume of the solid generated by revolving the triangular region bounded by the lines $y = 2x$, $y = 0$, and $x = 1$ about the line $x = 1$,

Answer: $\frac{8\pi}{6}$.

2. Solve the following integrals

a) $\int_0^1 \frac{e^x}{e^{2x}+4} dx$, **Answer:** $\frac{\arctan(\frac{e^x}{2})}{2} + C$

b) $\int_1^2 \frac{e^{\frac{1}{x}}}{x^2} dx$, **Answer:** $-e^{\frac{1}{x}} + C$.

c) $\int_3^9 x\sqrt{x+1}dx$, .
Answer: $40\sqrt{10} - \frac{20}{3}\sqrt{10} - \frac{64}{5} + \frac{16}{3}$

d) $\int \frac{dx}{\sqrt{x}+\sqrt[3]{x}}$, **Hint:** Perform a Substitution that changes these roots to powers.
Answer: $2\sqrt{x} - 3\sqrt[3]{x} + 6\sqrt[6]{x} - 6\ln(\sqrt[6]{x}+1) + C$.

e) $\int x^3\sqrt{1-x^2} dx$, **Answer:** $\frac{-(x-1)(x+1)^{\frac{3}{2}}(3x-2)}{15} + C$

f) $\int xe^{-x^2} dx$, **Answer:** $\frac{-e^{-x^2}}{2} + C$.

g) $\int_0^{\frac{\pi}{3}} (\cos x + \sec x)^2 dx$, **Answer:** $\frac{\sin 2x+10x}{4} + \tan x + C$

h) $\int_1^2 \frac{(\ln x)^2}{x^3} dx$
Answer: $-\frac{2\ln^2 x+2\ln x+1}{4x^2} + C$

i) $\int x \ln(1+x) dx$, **Answer:** $\frac{1}{2}(x^2-1)\ln(1+x) - \frac{1}{4}x^2 + \frac{1}{2}x + \frac{3}{4} + C$

j) $\int_0^\pi e^{\cos t} \sin 2t dt$,
Answer: $-2e^{\cos x}(\cos x - 1) + C$.

k) $\int_0^\pi 4 \sec^6 x \tan^6 x dx$, **Answer:** $4 \frac{63 \tan^{11} x + 154 \tan^9 x + 99 \tan^7 x}{693} + c$

l) $\int \tan^3 x \sec^3 x dx$, **Hint:** Use a similar technique as in the previous integral. **Answer:** $\frac{\sec^5 x}{5} - \frac{\sec^3 x}{3} + C$

m) $\int_0^{\frac{\pi}{2}} \frac{\cos t}{\sqrt{1+\sin^2 t}} dt$

n) $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \cot^2 x dx$, **Answer:** $\sqrt{3} - \frac{1}{3}\pi$.

$\tilde{n}$) $\int \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^5 x \csc^3 x dx$ **Answer:** $\frac{22}{105}\sqrt{2} - \frac{8}{105}$.

o) $\int_0^a x^2 \sqrt{a^2 - x^2} dx$, **Answer:** $\frac{1}{6} \sec^{-1}(x/3) - \sqrt{x^2 - 9}/(2x^2) + C$

p) $\int x\sqrt{1-x^4} dx$, **Answer:** $\frac{1}{4} \sin^{-1}(x^2) + \frac{1}{4} x^2 \sqrt{1-x^4} + C$.